

The mode less travelled: quantifying transport flexibility using agent-based models

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9th September 2026



Agenda

Motivations

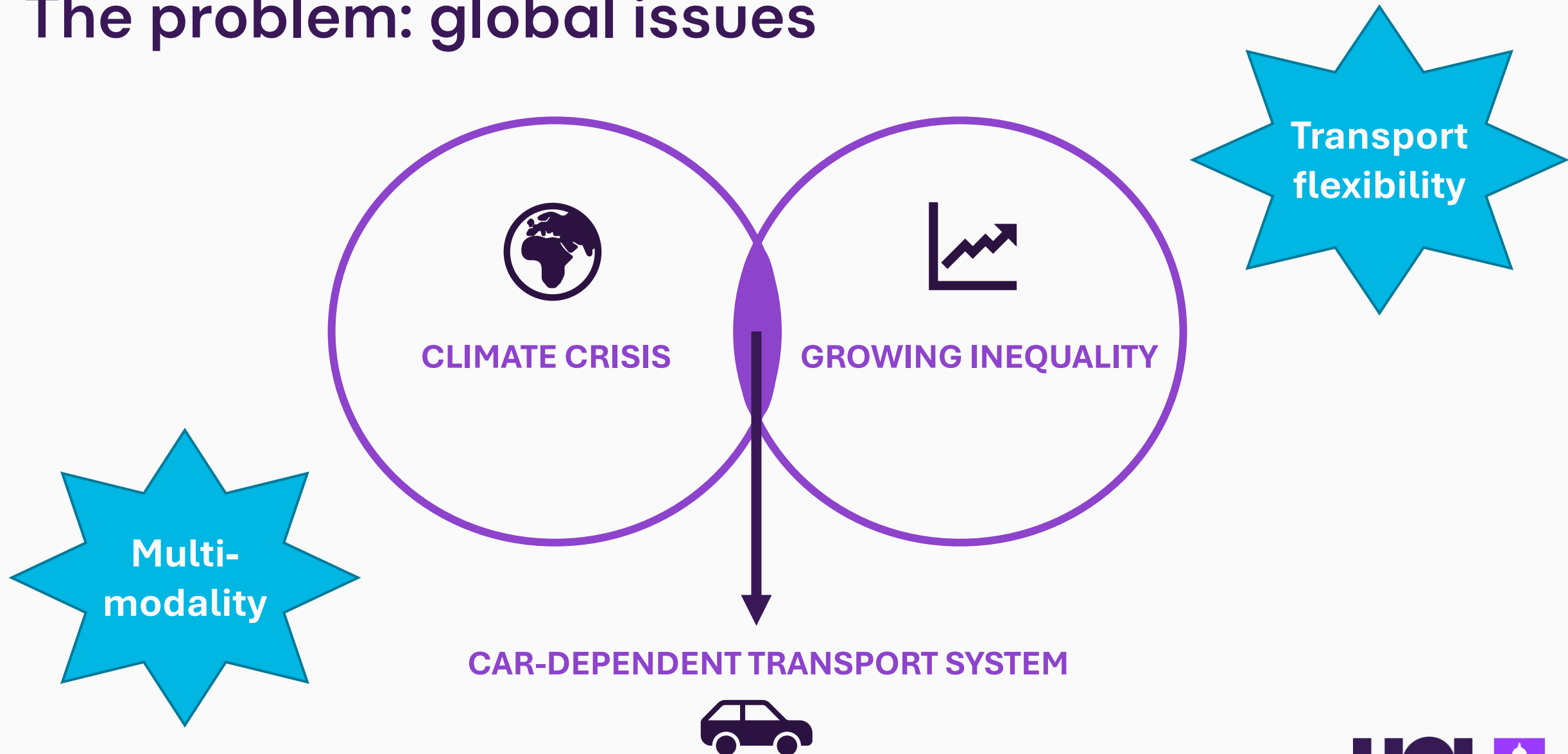
The concept of flexibility

Methods

Case study

Future work and conclusions

The problem: global issues



The concept: flexibility

Flexibility is an individual's ability to change their travel plans and still achieve their daily goals to the same standard

i.e., how much does changing travel plans affect your day?

Mode

Time

Route

Destination

Another problem: measurement vs reality



TRANSPORT PROVISION

“There is a bus stop
right outside my
house”



REALISABLE OPPORTUNITY

“The bus doesn’t go
where I need to go
though”

The gap is real, and is distributed unevenly across geographies and socio-demographic groups

The concept: simulation of alternatives

Agent-based simulation provides a unique perspective on not only what an agent **did** but also what they **could have done instead**

The alternatives are 'experienced' in a realistic **congested** and **dynamic** network

Selected plan: realized choice



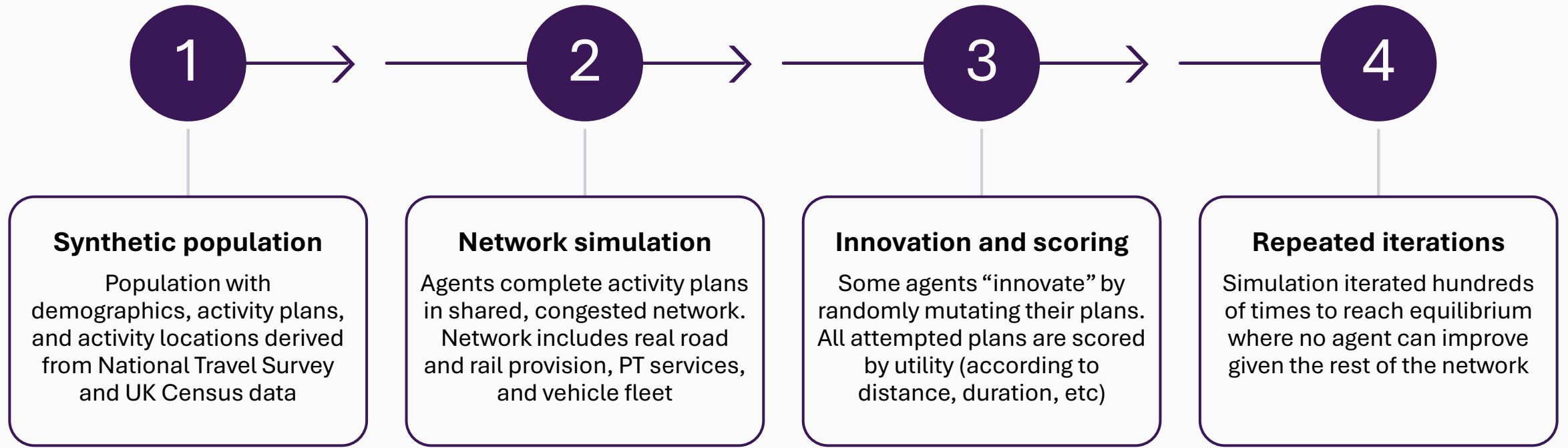
~ observed behaviour e.g., NTS

Unselected plans: potential, unrealized choices



~ stated preference in surveys – unreliable

MATSim in a nutshell

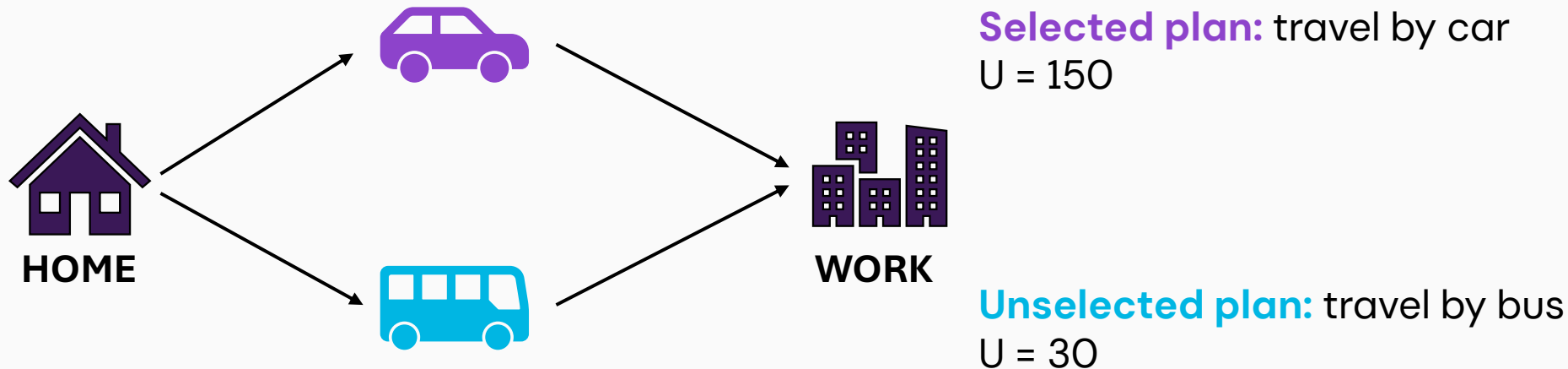


At equilibrium, all agents have a **selected plan** (what they did) and a set of **unselected alternative plans** (what they could have done). All plans have been **experienced** and **scored** in the shared network

The concept: simulation of alternatives

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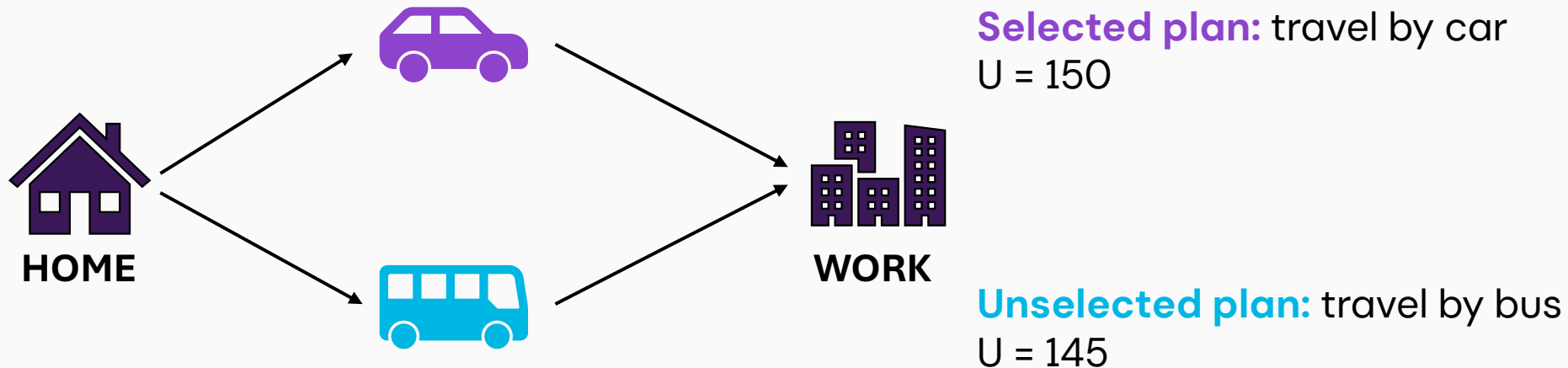


LOW FLEXIBILITY
HIGH COST OF MODE
SWITCH

The concept: simulation of alternatives

Agent-based simulation provides a unique perspective on not only what an agent **did** but also what they **could have done instead**

The alternatives are 'experienced' in a realistic **congested** and **dynamic** network



HIGH FLEXIBILITY
LOW COST OF MODE
SWITCH

Flexibility from MATSim outputs

Compare the selected plan (highest-scoring plan) to the best alternative that invokes a mode shift

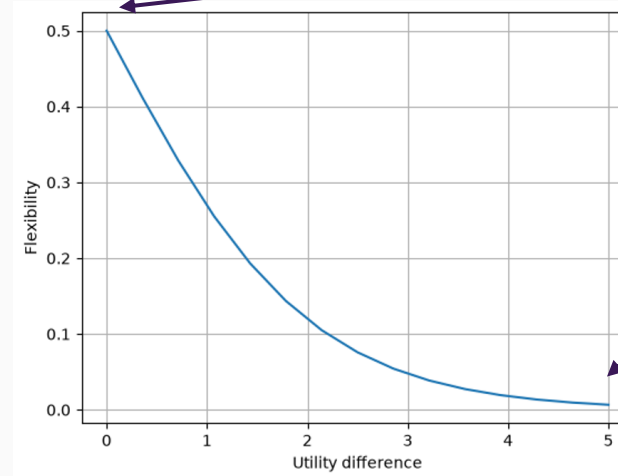
Transform the difference in utility using a **logistic function**

This gives the **binary logit choice probability** between these two plans in a head-to-head comparison

$$F = \frac{1}{1 + \exp(\mu \cdot \Delta U)}$$

$$\Delta U = U(\textit{selected}) - U(\textit{alternative})$$

$\mu = 1$ for consistency with MATSim



Perfect flexibility
Zero cost to mode switch

Near-zero flexibility
Mode switch cost
asymptotically large

Case study: car dependence in the East of England

Data: BERTIE (Behaviour and Transport: Impact and Equity)

- Developed by Transport East and Arup
- 10% representative synthetic population
- Activity and agent-based MATSim model of East of England

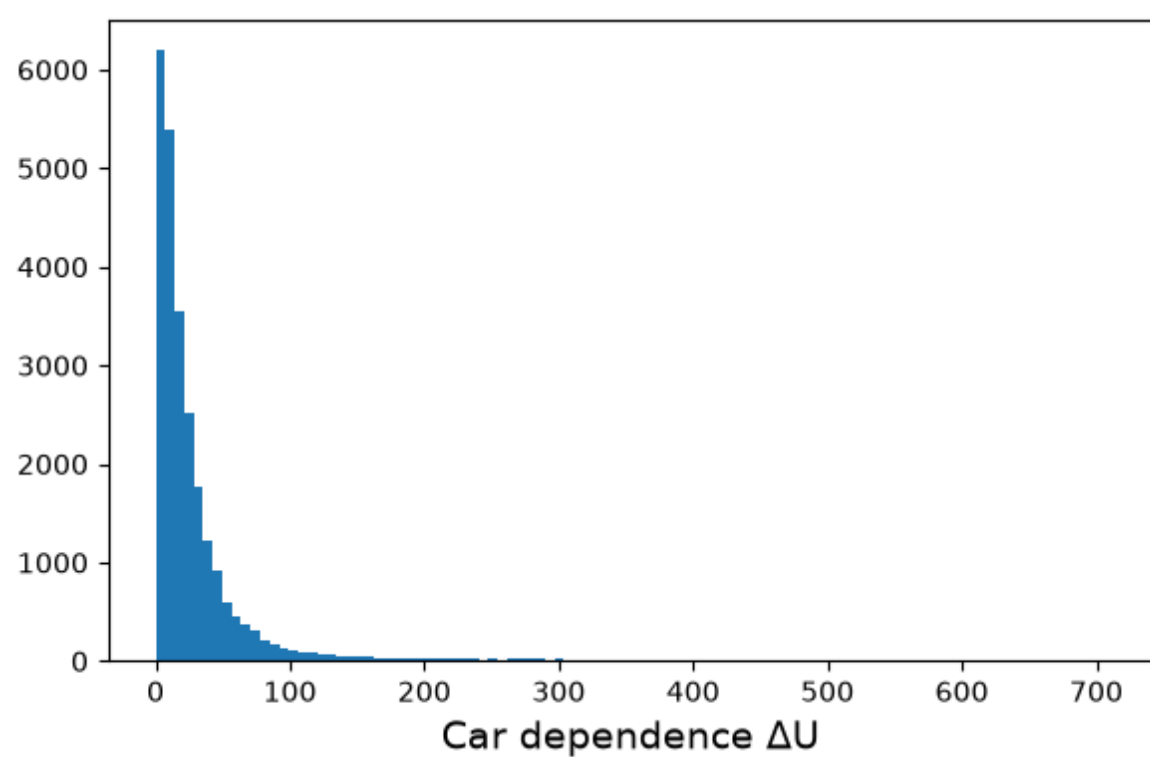


ARUP

RQ: To what extent can agents who travel by car reduce their car use?

- Select agents with car in selected plan (**57%**)
- Identify best alternative plan that reduces car use (by distance)
- Compare utilities and transform to find **F**
- Weight **F** by proportion of car reduction
- Validate results against DfT Connectivity (the “national measure of connectivity”)

Results: flexibility distribution



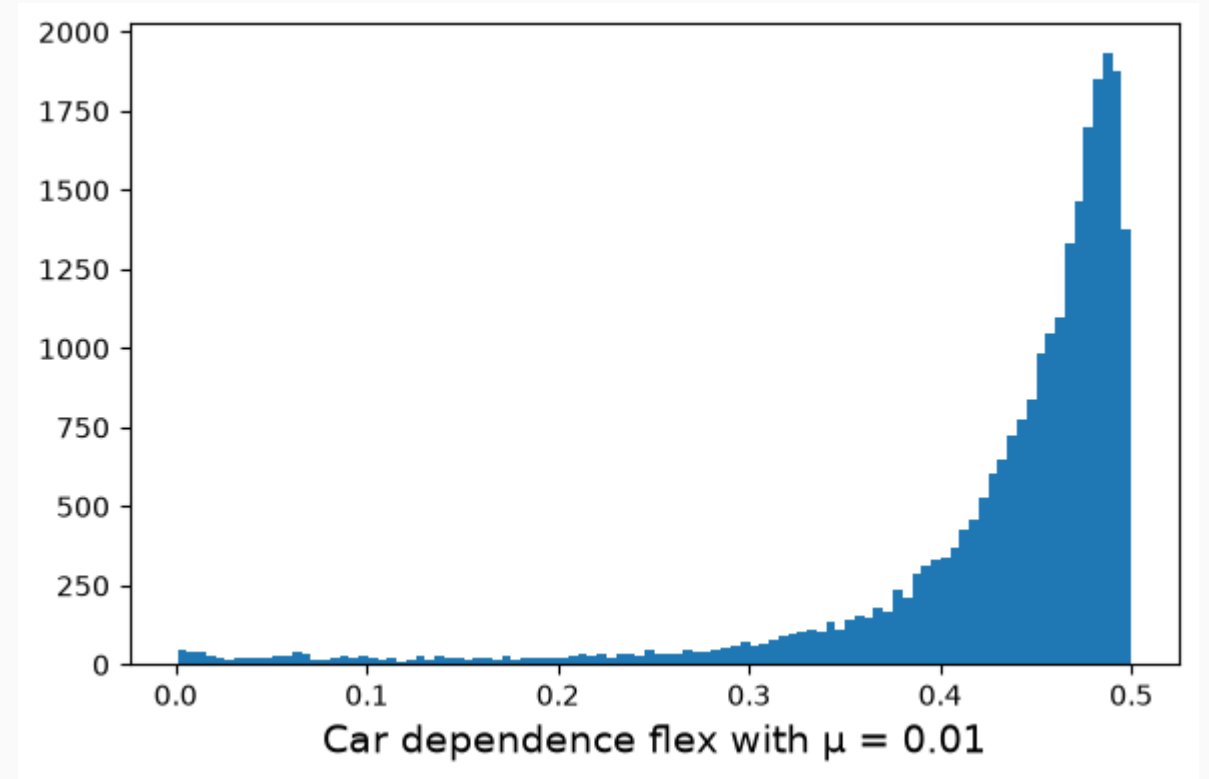
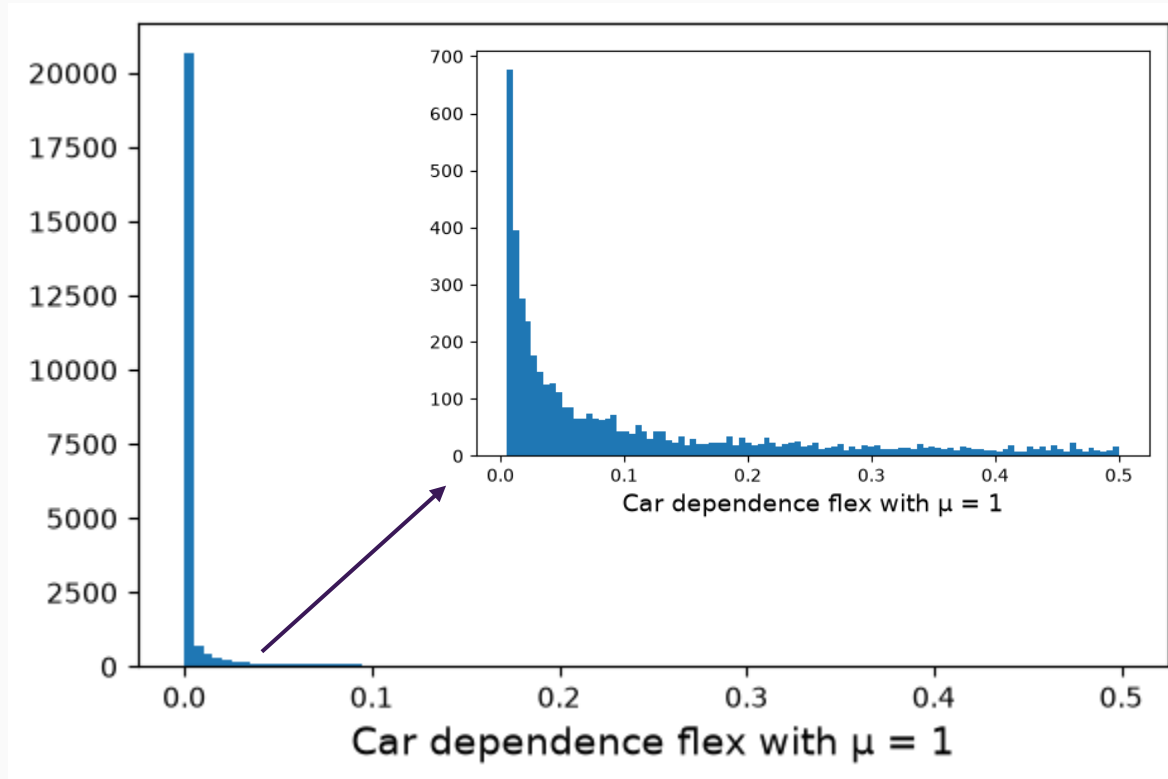
34% agents have $\Delta U < 10$ utils

- Many agents have a decent car-reducing alternative

5% agents have $\Delta U > 100$ utils

- Some agents have only very poor car-reducing alternatives

Results: flexibility distribution

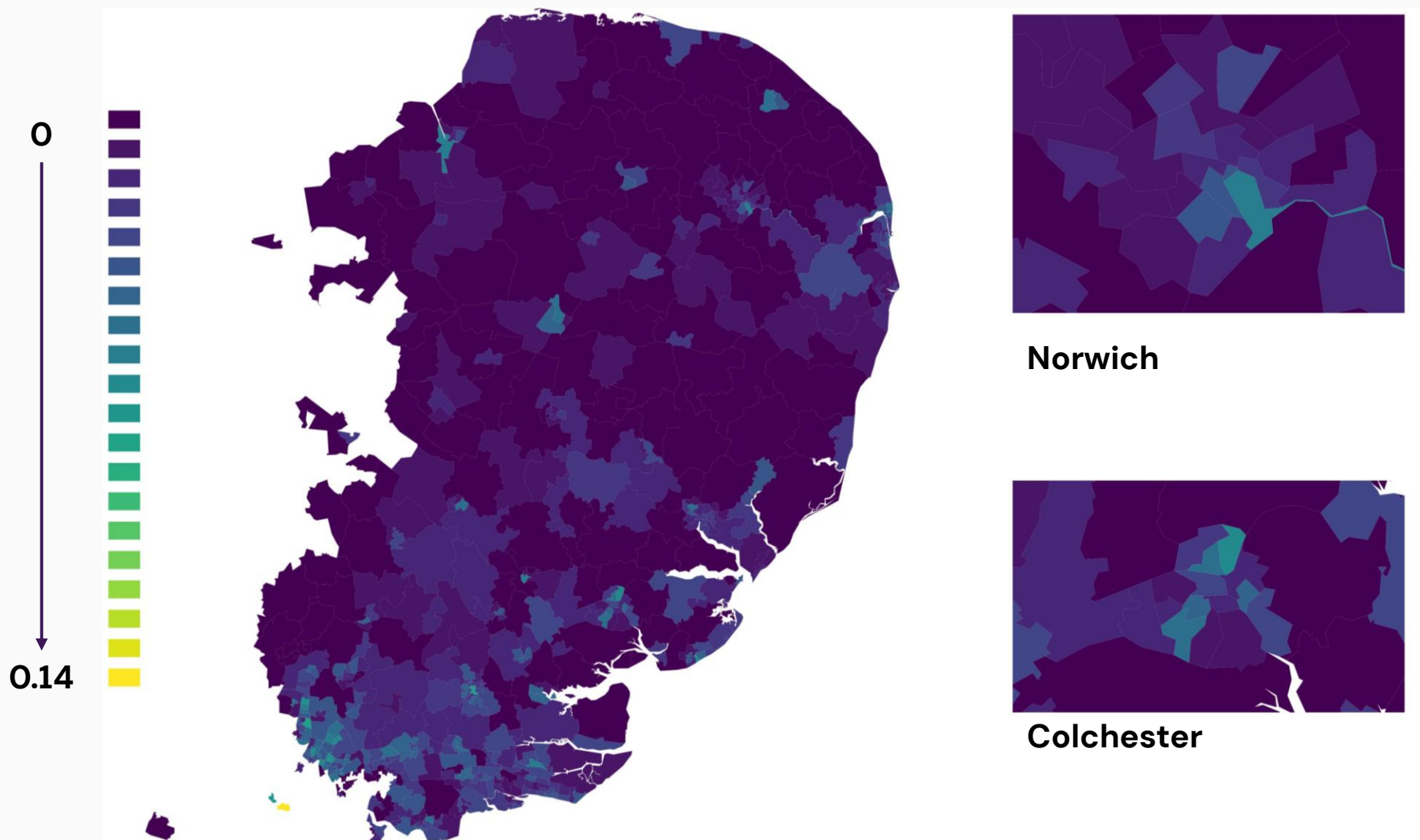


94% agents have $F < 0.05$ at $\mu = 1$

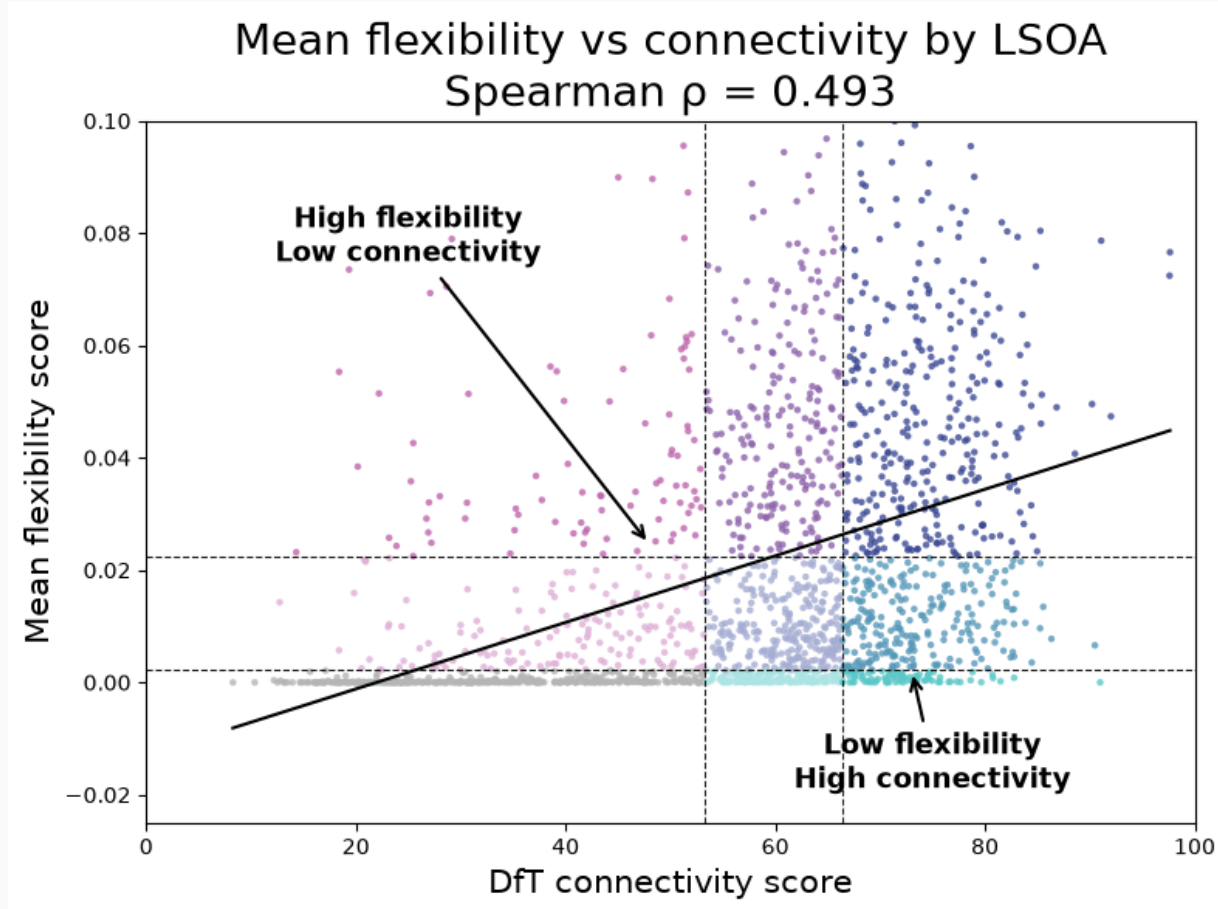
0.2% agents have $F > 0.45$ at $\mu = 1$

- Choice of μ influences shape of distribution

Results: geographical distribution



Results: comparison to connectivity



LSOAs in upper-left and lower-right quadrants show **decoupling**:

Resourceful mobility (3% of LSOAs)

High flexibility, low connectivity
Less deprived, lower PT stop density,
higher car ownership, semi-urban

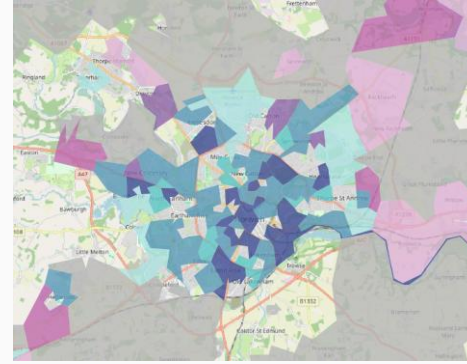
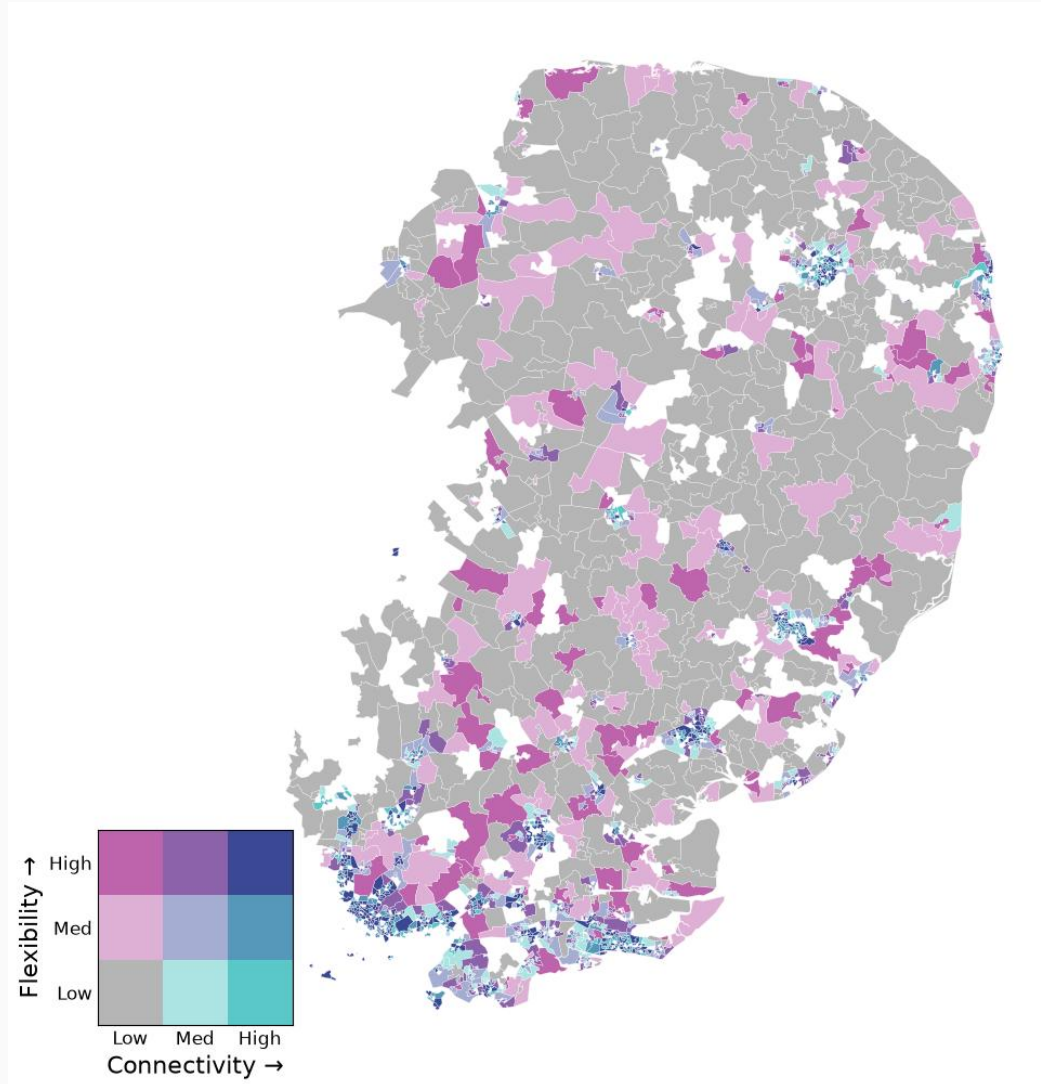
Unrealized provision (6% of LSOAs)

Low flexibility, high connectivity
More deprived, higher PT stop
density, lower car ownership, highly
urban

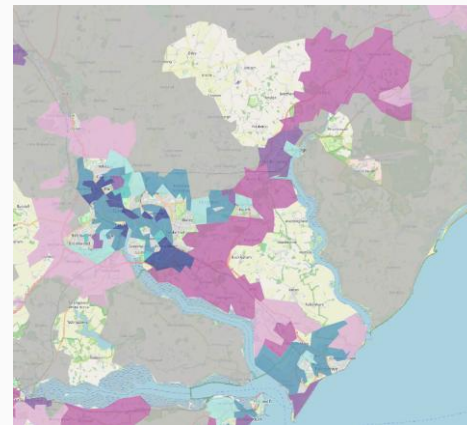
Spearman's $\rho = 0.5$

Moderate relationship but with much variance unexplained

Results: comparison to connectivity



Norwich



Ipswich + Woodbridge

Resourceful mobility (high flex, low conn)

Mostly periphery of urban centres

Unrealized provision (low flex, high conn)

Low flexibility, high connectivity

Mostly within urban centres

Extension: trip-level car dependence

Apply the methodology to a given trip in the day e.g., the **commute**
or school drop-off

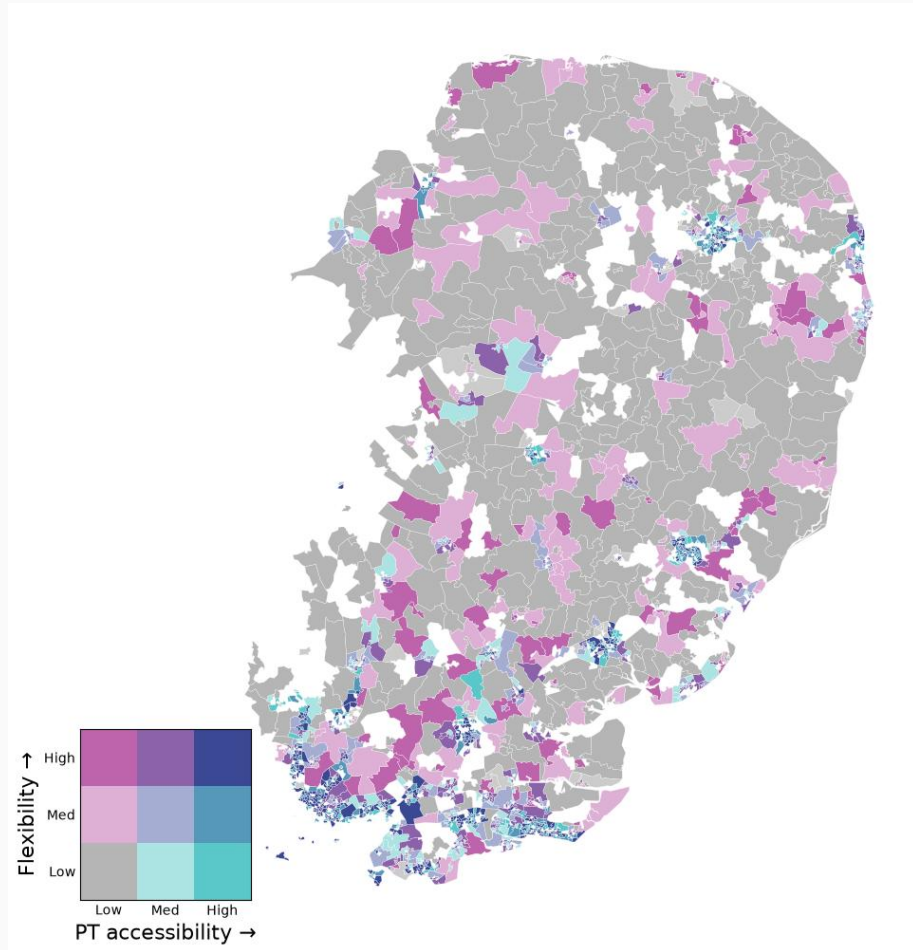
For agents who travel to work (or school dropoff) by car in their selected plan, compare the utility of this trip against alternative plans where car is not used

Include additional term to flexibility transform to account for **downstream scheduling effects**

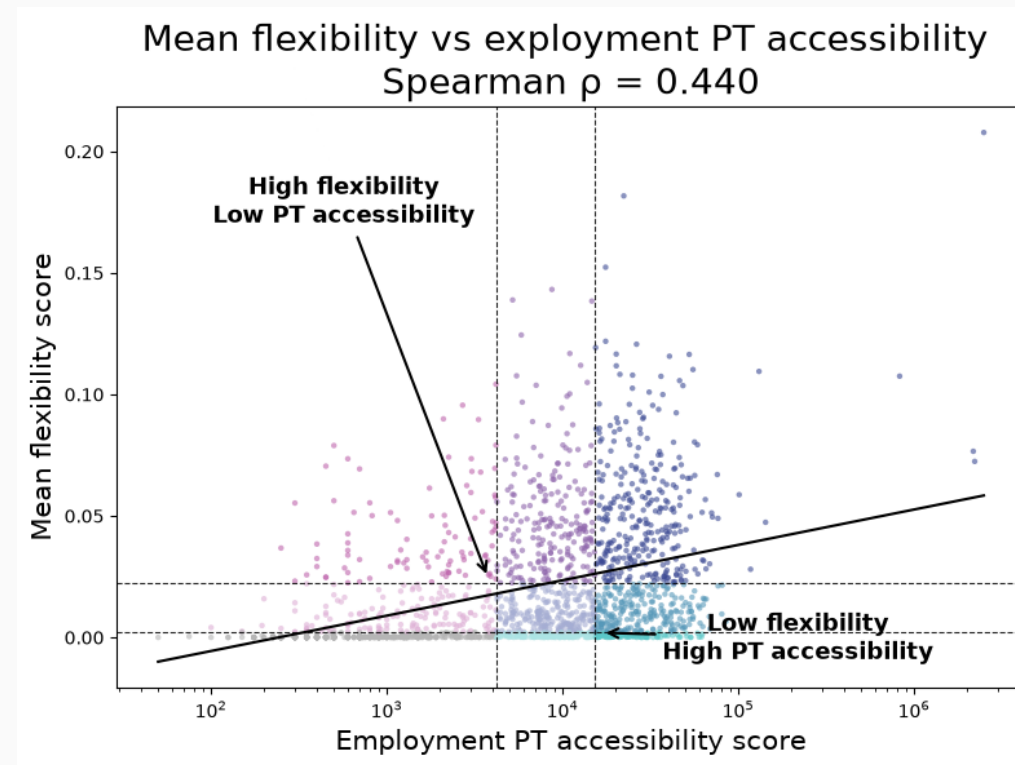
$$F = \frac{1}{1 + \exp(\mu \cdot (\Delta U_{trip} + \Delta U_{activity}))}$$

We can find which agents are able to reduce car across the day, but not for essential trips

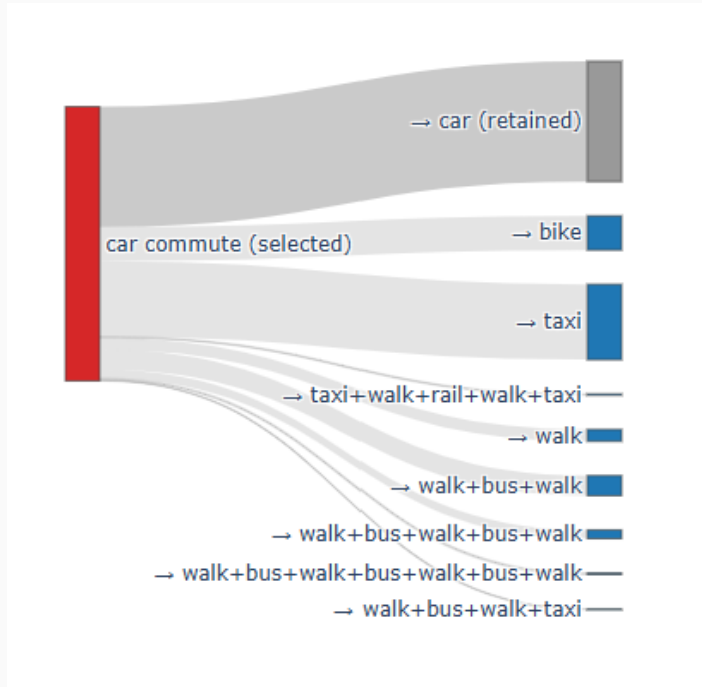
Extension: trip-level car dependence



Comparison to employment public transport accessibility indicators (30 minutes)

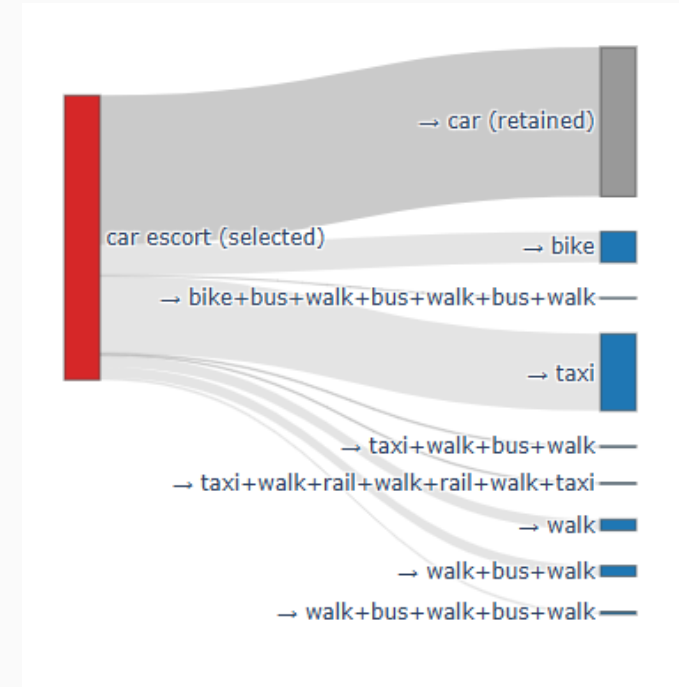


Extension: trip-level car dependence



Commute

42% of agents retain car in best day-level car-reducing alternative



School drop-off

49% of agents retain car in best day-level car-reducing alternative

Next steps

- Modal elimination (100% car reduction)
- Network-level modal resilience
- Disaster scenario testing including route flexibility
- Modelling of outcome distributions to understand relationships between demographics, geography, and flexibility more robustly
- Policy scenario testing for efficacy of interventions to improve flexibility

Conclusions

- Examining at unselected simulated travel plans can generate a novel quantified description of modal flexibility
- Flexibility can be assessed across an entire day or for a specific trip purpose
- Flexibility is not the same as connectivity – provision does not always translate into usefulness
- It's not just rural areas that experience low car-reducing flexibility
- Essential trips often hard to mode switch even in car-reducing alternatives

References

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Thank you!

Any questions?

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